Juan David Muñoz-Bolaños

Optical Metrology Engineer

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PROFESSIONAL SUMMARY

Optical Metrology Engineer with a specialized background in high-precision metrology, optical simulation, and Python/C++ automation. Proven expertise in building automated optical test setups from scratch and performing rigorous mathematical data analysis to guide technical design and system integration.

TECHNICAL COMPETENCIES

Optical Metrology & Simulation
Expert in wavefront sensing, phase retrieval, and ZEMAX OpticStudio tolerancing. Proficient in characterizing aberration sensitivity of high-NA microscope objectives.

System Automation & Control
Experience building C++/Python/LabVIEW control stacks for adaptive optics, scientific cameras (sCMOS), and opto-mechanical stages.

Data Science for Optics
Application of supervised ML and AI algorithms to hyperspectral imaging data. Proficient with NumPy, Pandas, Scikit-learn, and JAX for high-performance computation.

Hardware Integration & Sourcing
End-to-end R&D experience in multi-photon microscope assembly. Competent in technical component specification (Thorlabs, Edmund Optics) and incoming QA.

Process Reliability
Hands-on experience with industrial vision solutions (Halcon) and ensuring measurement traceability via Gage R&R assessments.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck — Innsbruck, Austria
PhD Researcher - Optical Metrology & Automation — *Jan 2022 – Present*

- **Automated Metrology Setups:** Designed and built an automated wavefront sensing system; developed the full software control stack to unify scientific cameras and wavefront modulators.
- **Optical Modeling & Design:** Performed systematic ZEMAX tolerance analysis to optimize the process of custom objective lens characterization and aberrometry.
- **Integration & Alignment:** Achieved diffraction-limited performance in a custom-built nonlinear microscope by precisely aligning galvo scanners, femtosecond lasers, and PMT detectors.
- **Technical Documentation:** Managed technical specification sheets and qualified vendors (Hammamatsu, Thorlabs) for high-precision opto-mechanical integration.

Leibniz Institute of Photonic Technology (IPHT) — Jena, Germany
R&D Researcher (Photonics & Data) — *Jun 2020 – Dec 2021*

- **Spectrometer Build:** Handled the end-to-end optical layout and detector calibration of a 1064 nm fiber-probe Raman spectrometer prototype.
- **AI-Driven Data Analysis:** Leveraged custom Python packages for Machine Learning classification on hyperspectral tissue images, accelerating experimental turnaround.
- **Published Research:** Contributed to the development of non-invasive clinical diagnostic tools, results documented in peer-reviewed journals (Applied Sciences, ACS Photonics).

INTECOL S.A.S. — Medellín, Colombia

Junior Machine Vision System Engineer *Mar 2018 – Mar 2019*

- **Industrial Machine Vision:** Engineered Halcon inspection systems for beverage production lines; solved critical label and sealing quality issues via custom illumination design.
- **Hardware Commissioning:** Optimized robotic vision for auto-part recognition using Cognex VisionPro, directly impacting production efficiency at client sites like Renault.
- **Gage R&R Assessments:** Verified system accuracy and process stability within industrial quality control frameworks.

EDUCATION

Medical University of Innsbruck Austria
Ph.D. in Adaptive Optics & Microscopy *Jan 2022 – Present*

Friedrich Schiller University Jena Germany
M.Sc. in Photonics *2019 – 2021*

University of Cauca Colombia
B.Sc. in Physics Engineering *2012 – 2018*

PUBLICATIONS & AWARDS

- **Awards:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte Medical University of Innsbruck
Head of Institute for Biomedical Physics
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Dr. Christoph Krafft Leibniz Institute of Photonic Technology
Group leader: Spectroscopy Imaging
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